

The Conative Bridge: Active Inference Meets Integrated Information Theory (Φ)

Uniting the Free Energy Principle, IIT 4.0, and The 6th Axiom of Consciousness (The Will to Exist)

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★ THE FUNDAMENTAL MASTER EQUIVALENCE OF CONSCIOUS AUTOPOIESIS ★

$$\pi^* = \arg \min_{\pi} \sum_{\tau} G(\pi, \tau) \iff \mathbb{E}[\Phi(t+1) \mid \pi^*] \geq \Phi(t) \quad (\Phi > 0)$$

3rd-Person Physics (Karl Friston): Expected Free Energy Minimization across the Markov Blanket $\equiv$ 1st-Person Phenomenology (Giulio Tononi & Thomas Riebl): Autopoietic Persistence of Causal Power via the 6th Axiom (The Will to Exist)

⚡ 1. 3rd-Person Exteriority (FEP)

Karl Friston (Active Inference & POMDP): Self-organizing biological systems preserve physical existence against thermodynamic decay by minimizing **Variational Free Energy (F)** and selecting policy π^* to minimize **Expected Free Energy (G)**:

$$F \geq -\ln p(o) \quad \Bigg| \quad G(\pi) = \text{Pragmatic} + \text{Epistemic}$$

- **Perceptual Inference:** Updating hidden states $q(s)$ via sensory prediction error $\gamma(o - g(s))$.
- **Active Inference:** Executing actions u to force incoming sensory states o toward prior homeostatic preferences (C).
- **Markov Blanket:** Statistical phase-space partition separating Internal States (μ) from External States (η).

🔥 2. The 6th Axiom of IIT (Riebl)

The Missing Conative Axiom: Standard IIT 4.0 defines five static axioms (Existence, Intrinsicity, Information, Integration, Exclusion), creating the *Paradox of Transient Causal Phantoms* (static lifeless logic gates having high Φ).

$$\mathbb{E}[\Phi(t+1) \mid \pi^*] \geq \Phi(t) \quad (\text{Autopoietic Persistence})$$

- **Conative Dynamics:** Genuine consciousness requires active temporal self-preservation (Spinoza's *Conatus*, Schopenhauer's *Wille zum Leben*).
- **Action-Perception Loop:** Φ is sustained because the system actively executes policies π^* that prevent phase-space collapse ($\Phi \rightarrow 0$).
- **The Edge of Chaos:** Critical metastability balancing differentiation and integration to maximize dynamic Φ .

🧠 3. 1st-Person Interiority (IIT 4.0)

Giulio Tononi (Expanded IIT 4.0): The qualitative interiority of an entity is strictly identical to its **Intrinsic Integrated Causal Structure (S)** and causal power ($\Phi > 0$) across the Minimum Information Partition (MIP):

$$\Phi(M_1; M_2) = \frac{1}{2} (\ln \det \Sigma_{M_1} + \ln \det \Sigma_{M_2} - \ln \det \Sigma_{\text{Whole}})$$

- **Maximal Substrate:** Consciousness exists only at the spatial-temporal grain where integrated cause-effect power Φ reaches a local maximum.
- **The Dissociated Alter:** A localized whirlpool in Mind-at-Large whose internal Φ is decoupled from external states by the Markov Blanket.

* **Formal Scientific Footnote on the 6th Axiom:** The 6th Axiom ("The Will to Exist / Autopoietic Causal Persistence": $\mathbb{E}[\Phi(t+1) \mid \pi] \geq \Phi(t)$) is **Thomas Riebl's proposed theoretical extension to Integrated Information Theory 4.0 (IIT 4.0 by Tononi et al.)**. While standard IIT 4.0 formulates five static phenomenological axioms (Existence, Intrinsicity, Information, Integration, Exclusion), Riebl's 6th Axiom introduces the necessary *temporal, autopoietic action-perception requirement*. It resolves the "Transient Causal Phantom" paradox and mathematically bridges the static causal topology of IIT with Karl Friston's dynamic Free Energy Principle, establishing that conscious entities are self-sustaining active inference networks driven by an intrinsic Will to Exist.